

Semidefinite Programming in Quantum Information

Lecture 1: Notation, SDPs, Duality, and Canonical Examples
Quantum in the Dunes

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Scope & takeaways

By the end of the hour, you should be able to:

- Recognize SDPs as “quantum-feasible” optimization problems (PSD constraints everywhere).
- Write a QI task as an SDP (states / POVMs / simple operator inequalities).
- Derive the **dual** using a standard Lagrangian template.
- Use the dual as a **certificate**: bounds, witnesses, and optimality checks.

Main storyline

(QI objects) $\Rightarrow$ (convex feasible sets) $\Rightarrow$ (SDPs) $\Rightarrow$ (duality and certificates).

QI linear algebra: PSD cone, states, and measurements

Hermitian and PSD

- $X \in L(\mathcal{X})$ is **Hermitian** if $X = X^*$.
- $X \succeq 0$ (**PSD**) iff $\langle v, Xv \rangle \geq 0$ for all v .
- PSD $\Rightarrow$ eigenvalues ≥ 0 .

States and POVMs

- **State:**
 $\rho \in D(\mathcal{X}) = \{\rho \succeq 0 : \text{Tr}(\rho) = 1\}$.
- **POVM:** $\{M_a\}_a$ with $M_a \succeq 0$,
 $\sum_a M_a = \mathbb{1}$.
- Born rule: $\text{Pr}(a) = \text{Tr}(M_a \rho)$.

Inner product (workhorse)

$$\langle X, Y \rangle = \text{Tr}(X^* Y) \quad (\text{for Hermitian: } \langle X, Y \rangle = \text{Tr}(XY)).$$

Convexity is the reason optimization behaves well

Convexity (1 line)

A set C is convex if $tx + (1 - t)y \in C$ for all $x, y \in C$, $t \in [0, 1]$.

Key QI fact

$D(\mathcal{X})$ is convex:

$$\rho_1, \rho_2 \in D(\mathcal{X}) \Rightarrow t\rho_1 + (1 - t)\rho_2 \in D(\mathcal{X}).$$

Same for POVMs (componentwise PSD + affine sum constraint).

Why we care

- Many operational quantities are linear in ρ or M_a .
- Linear objective + convex constraints $\Rightarrow$ **no local traps**.
- Duality gives **certificates** (bounds, witnesses).

Linear maps and adjoints (duals)

Adjoint map

For $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y})$, the adjoint $\Phi^* : L(\mathcal{Y}) \rightarrow L(\mathcal{X})$ satisfies

$$\langle Y, \Phi(X) \rangle = \langle \Phi^*(Y), X \rangle \quad \forall X, Y.$$

Completely positive maps

Φ is CP if $\Phi \otimes \text{id}_{\mathcal{Z}}$ is positive for all $\mathcal{Z}$. (Channels are CPTP; Choi matrices make CP a PSD constraint. All SDP-friendly.)

Why SDPs keep showing up in quantum information

A recurring template

Optimize a linear functional over states/POVMs/channels (convex + PSD constraints).

Where this appears

- state discrimination / hypothesis testing
- entanglement witnesses and monotones (relaxations)
- norm bounds (trace norm / operator norm)
- channel optimization (via Choi constraints)

What you get from duality

- upper/lower bounds “for free”
- interpretability: witnesses, inequalities, converse bounds
- optimality proofs via [matching primal/dual](#)
- structural insights via complementary slackness

What is an SDP? (LP with matrix positivity)

High-level definition

An SDP optimizes a **linear** function of a Hermitian matrix variable X subject to:

- linear equalities/inequalities in X ,
- one nonlinear-looking constraint: $X \succeq 0$ (sometimes Hermitian preserving map applied to X too).

Classical analogy

$$\text{LP: } x \geq 0 \quad \rightsquigarrow \quad \text{SDP: } X \succeq 0.$$

QI intuition

$\rho \succeq 0$ and $M_a \succeq 0$ are physical constraints coming from quantum mechanics!

Standard primal/dual forms

Primal (P)

maximize: $\langle C, X \rangle$
subject to: $\langle A_i, X \rangle = b_i \quad (i = 1, \dots, m)$
 $X \succeq 0$.

$X \in \text{Herm}(\mathcal{X})$; $C, A_i \in \text{Herm}(\mathcal{X})$; $b \in \mathbb{R}^m$.

Dual (D)

minimize: $b^\top y$
subject to: $\sum_{i=1}^m y_i A_i \succeq C$
 $y \in \mathbb{R}^m$.

Philosophy

Dual feasible $\Rightarrow$ certificate (bound) on primal optimum. The topic of this lecture!
SDPs can (almost always) be solved in polynomial time.

Example 1: largest eigenvalue as an SDP

Let $M \in \text{Herm}(\mathcal{X})$. Variational principle:

$$\lambda_{\max}(M) = \max_{\rho \in \mathcal{D}(\mathcal{X})} \text{Tr}(M\rho).$$

Primal

maximize: $\text{Tr}(M\rho)$
subject to: $\text{Tr}(\rho) = 1$
 $\rho \succeq 0$.

Dual

minimize: λ
subject to: $\lambda \mathbb{1} \succeq M$.

Takeaway

Dual says: $\lambda_{\max}(M) = \min\{\lambda : \lambda \mathbb{1} \succeq M\}$.

Example 2: minimum-error state discrimination (primal)

Ensemble $\mathcal{E} = \{(p_a, \rho_a)\}_{a=1}^k$, $\rho_a \in \mathcal{D}(\mathcal{X})$.

Optimization problem

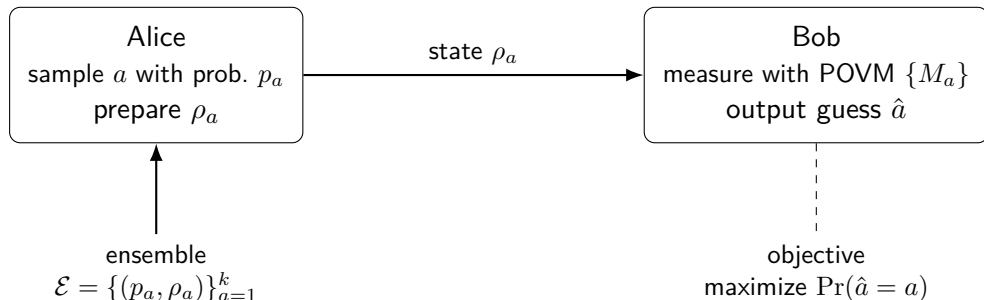
Choose a POVM $\{M_a\}_{a=1}^k$ to maximize success probability

$$p_{\text{succ}} = \sum_{a=1}^k p_a \text{Tr}(M_a \rho_a).$$

SDP (primal)

$$\begin{aligned} &\text{maximize: } \sum_{a=1}^k p_a \text{Tr}(M_a \rho_a) \\ &\text{subject to: } \sum_{a=1}^k M_a = \mathbb{1} \\ &\quad M_a \succeq 0 \quad (a = 1, \dots, k). \end{aligned}$$

Operational picture (why this SDP is the right model)



Born rule $\Rightarrow$ objective

$$\Pr(\hat{a} = a) = \sum_{a=1}^k p_a \Pr(\hat{a} = a \mid a) = \sum_{a=1}^k p_a \text{Tr}(M_a \rho_a).$$

State discrimination: the dual and its meaning

Dual SDP

$$\begin{aligned} & \text{minimize: } \text{Tr}(H) \\ & \text{subject to: } H \succeq p_a \rho_a \quad (a = 1, \dots, k), \\ & \quad \quad \quad H \in \text{Herm}(\mathcal{X}). \end{aligned}$$

Interpretation (certificate view)

Any feasible H upper-bounds each $p_a \rho_a$ in semidefinite order, hence

$$p_{\text{succ}}^* \leq \text{Tr}(H) \quad (\text{by weak duality}).$$

When strong duality holds, the best such bound is tight.

What to remember

Primal = best measurement. Dual = best operator upper bound (a “witness”).

Deriving the dual: the Lagrangian template

Step 1: Lagrangian

Introduce multipliers $y \in \mathbb{R}^m$ for equalities and slack $S \succeq 0$ for $X \succeq 0$:

$$\mathcal{L}(X, y, S) = \langle C, X \rangle + \sum_{i=1}^m y_i (b_i - \langle A_i, X \rangle) - \langle S, X \rangle.$$

Rearrange:

$$\mathcal{L}(X, y, S) = b^\top y + \left\langle C - \sum_i y_i A_i - S, X \right\rangle.$$

Step 2: boundedness forces a constraint

If $C - \sum_i y_i A_i - S \neq 0$, then $\sup_X \mathcal{L}(X, y, S) = +\infty$ (scale X). Hence we must have

$$C - \sum_i y_i A_i - S = 0 \iff \sum_i y_i A_i \succeq C.$$

Duality as “automatic inequalities” (weak duality)

Proposition (Weak duality)

For any primal feasible X and dual feasible y ,

$$\langle C, X \rangle \leq b^\top y.$$

One-line proof idea

Dual feasibility gives $S := \sum_i y_i A_i - C \succeq 0$. Then for $X \succeq 0$,

$$0 \leq \langle S, X \rangle = \sum_i y_i \langle A_i, X \rangle - \langle C, X \rangle = b^\top y - \langle C, X \rangle.$$

Interpretation

Every dual feasible point gives a valid upper bound on the primal optimum.

Strong duality: Slater's condition (statement only)

Theorem (Slater (informal but standard))

*If the primal is feasible and has a **strictly feasible** point $X_0 \succ 0$ satisfying the affine constraints, then:*

- *there is no duality gap: $\max(\mathbf{P}) = \min(\mathbf{D})$,*
- *the dual optimum is attained.*

Similarly with strict feasibility on the dual side.

QI sanity checks (practical Slater)

- States: $\rho = \mathbb{1}/d$ is full rank.
- POVMs: $M_a = \mathbb{1}/k$ is full rank (each outcome strictly PSD).
- Often: add $\varepsilon \mathbb{1}$ (regularization) to get strict feasibility.

A warning slide: duality gap can happen without Slater

Pathology (one-liner)

Without strict feasibility, strong duality can fail: finite primal optimum, infeasible dual (gap). There is an explicit example in the lecture notes.

Why you should care in QI

- Many papers implicitly “swap primal and dual” in proofs. But do not check it!
- This is safe only under a condition like Slater.
- In QI models, strict feasibility is often easy to ensure (full rank / $\varepsilon \mathbb{1}$).
- Technical note: most solver also need some bound on the feasible set beyond strong duality. I.e. the promise that the trace of primal/dual solutions is upper-bounded by something at least exponential.

Complementary slackness: structure and optimality checks

Setup

Dual slack: $S := \mathcal{A}^*(y) - C \succeq 0$.

Theorem (Complementary slackness (SDPs))

Assume strong duality holds and X^*, y^* are optimal. With

$$S^* := \mathcal{A}^*(y^*) - C \succeq 0,$$

we have

$$\text{Tr}(S^* X^*) = 0 \quad \Rightarrow \quad S^* X^* = 0 \quad \text{and} \quad \text{supp}(X^*) \subseteq \ker(S^*).$$

How you actually use this

- **Certify optimality:** find feasible X, y with $\text{Tr}(SX) = 0$.
- **Extract structure:** optimal solution lives in the kernel of dual slack.

A reusable recipe (the slide you want in your head and where we are going next)

Modeling checklist (QI $\rightarrow$ SDP)

- Variables are operators: ρ , M_a , Choi matrices, moment matrices, ...
- Physical constraints are affine + PSD: $\rho \succeq 0$, $\text{Tr}\rho = 1$; $M_a \succeq 0$, $\sum_a M_a = \mathbb{1}$.
- Objective is typically linear: $\text{Tr}(H\rho)$ or $\sum_a \text{Tr}(M_a X_a)$.

Duality checklist (SDP $\rightarrow$ certificate)

- Dual feasibility $\Rightarrow$ bound (weak duality).
- Slater $\Rightarrow$ bound is tight + optimum attained.
- Complementary slackness $\Rightarrow$ optimality/structure.

Further reading & next steps

References (good starting points)

- Watrous-style lecture notes on SDPs in QI (dual interpretations are especially clean).
- Standard convex optimization texts for Slater/duality/interior-point background.

Where we go next

- Lecture 2: more QI examples (hypothesis testing, norms, channel constraints).
- Lecture 3: how to solve SDPs (modeling + solver behavior).

One meta-takeaway: in QI, dual variables are often the *right* objects.